Oct. 11

Test next Fri.

Please try to arrive 5 min. early.

One side 8.5×11" sheet for notes (no examples)

Starts with Taylor series → separation of variables

HW #9 sol'ns will be posted on Tues.

S'

Ch 7 § 5 More on Properties of Sol'n to Laplace Eq'n

property of harmonic fns

Recall: $u \in C^0(\Omega)$ $\Delta u = 0 \Rightarrow u \in C^\infty(\Omega)$

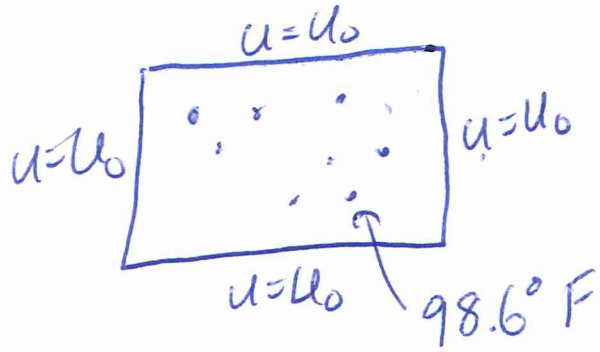
Physical interp of $\Delta u = 0$: steady-state temp. distribution

(Also $\Delta c = 0 \Rightarrow$ steady-state concentration of solute in fluid)

$u \in C^2(\Omega) \cap C^0(\bar{\Omega})$

ex) $\Delta u = 0$ Ω

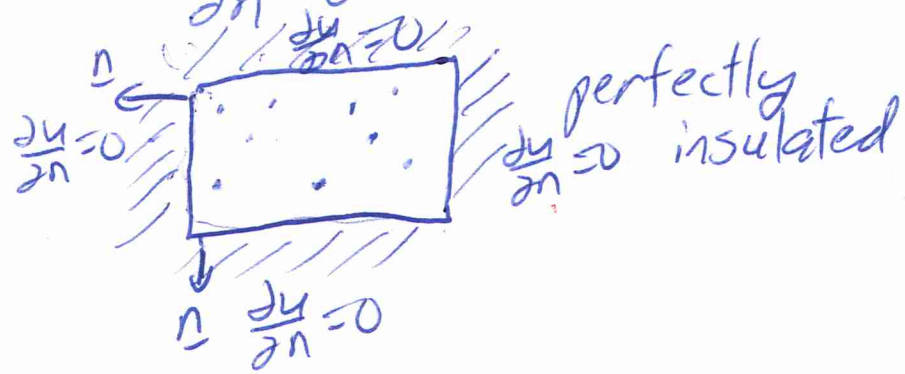
$u = u_0$ $\partial\Omega$



$\Rightarrow u = u_0$ in $\bar{\Omega}$

$\Delta u = 0$ Ω

$\frac{\partial u}{\partial n} = 0$ $\partial\Omega$



$u = C$ $C = ?$

Property 2 of solns to $\Delta u = 0$ (harmonic fns)

Thm <u>Mean-value property (MVP)</u>

If u is harmonic in the domain $\Omega \subset \mathbb{R}^n$,

Then u has the mean-value property

i.e.

$$u(\underline{r}_0) = \frac{1}{|S(\underline{r}_0, \delta)|} \int_{S(\underline{r}_0, \delta)} u(\underline{r})\, d\sigma$$

for any pt. $\underline{r}_0 \in \Omega$ and $\delta > 0$ where $B(\underline{r}_0, \delta) \subset \Omega$ and $S(r_0, \delta) = \partial B(r_0, \delta)$

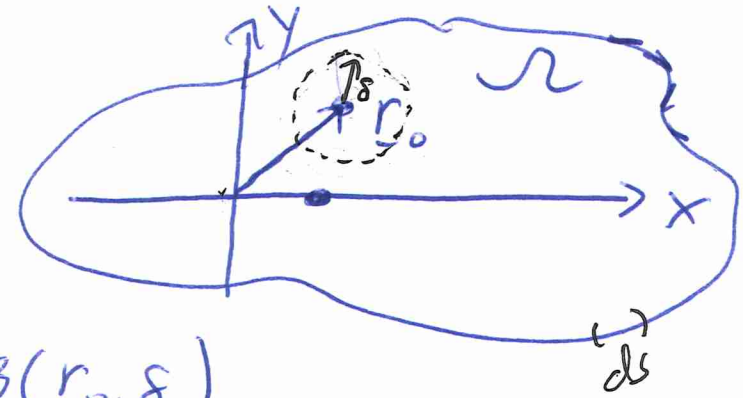
know

$|S(\underline{r}_0, \delta)|$ = surface area.

In $\mathbb{R}^2$: $B(\underline{r}_0, \delta)$

$\{\underline{x} \in \mathbb{R}^2 : |\underline{x} - \underline{r}_0| < \delta\}$

dist



$S(\underline{r}_0, \delta)$ = bdry of $B(\underline{r}_0, \delta)$

$= \{\underline{x} \in \mathbb{R}^2 : |\underline{x} - \underline{r}_0| = \delta\}$

$ds = \delta d\theta$

$|S(\underline{r}_0, \delta)| = 2\pi\delta$

$$u(\underline{r}_0) = \frac{1}{2\pi\delta} \int_{\theta=0}^{2\pi} u(\underline{r} = \underline{r}_0 + \underline{\delta}, \theta)\, \underbrace{\delta}\, d\theta$$

$r = \cancel{r_0}$

In $\mathbb{R}^3$ $B(\underline{r}_0, \delta) = \{\underline{x} \in \mathbb{R}^3 : |\underline{x} - \underline{r}_0| < \delta\}$

dist

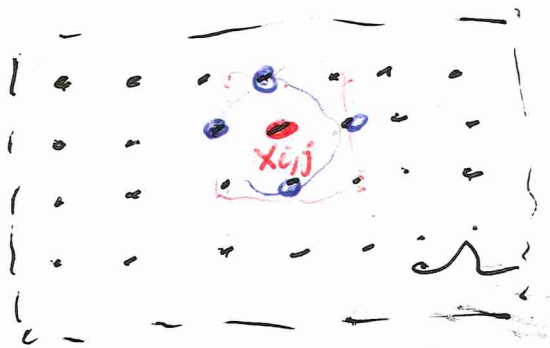
$S(\underline{r}_0, \delta) = \{\underline{x} \in \mathbb{R}^3 : |\underline{x} - \underline{r}_0| = \delta\}$

$|S(\underline{r}_0, \delta)| = \cancel{\frac{4}{3}\pi}\ 4\pi\delta^2$

$V = \frac{4}{3}\pi\delta^3$

$dS = \delta^2 \sin\phi\, d\theta\, d\phi$

$\Rightarrow$ Numerically solving $\Delta u = 0$ (not resp. for)



$$u_{(i,j)} = \cancel{\tfrac{1}{8}u}$$

$$= \frac{1}{4}\left[u_{(i+1,j)} + u_{(i-1,j)} + u_{(i,j+1)} + u_{(i,j-1)}\right]$$

Property 3 of harmonic fns

Thm 5.4 (Maximum Principle)

Let Ω be a bounded domain in $\mathbb{R}^n$ and suppose $u \in C^0(\bar{\Omega})$ and u is harmonic in Ω. Then u obtains its max & min values on the boundary $\partial\Omega$ and only on $\partial\Omega$, unless u is constant.

Pf ~~Maybe~~ We will show max. property. } Not responsible for.
Min. property is shown similarly

Suppose that $\underline{r}_0 \in \Omega$ ($\underline{r}_0 \notin \partial\Omega$) is such that $u(\underline{r}_0)$ is max. We will show that $u(\underline{r}_0) = \text{const.}$

By MVP

$$u(\underline{r}_0) = M = \left(\frac{1}{|S^{4\pi}\delta^2|}\right) \int_{S(\underline{r}_0,\delta)} u(\underline{r})\, d\sigma$$

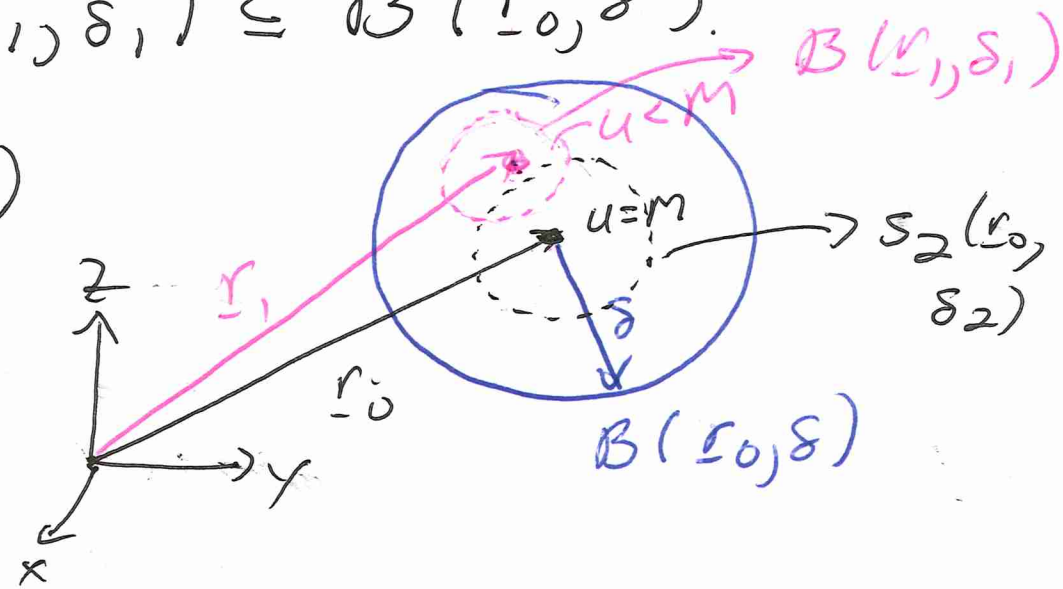
$\Rightarrow$ in $\mathbb{R}^3$

for δ small enough

<u>Claim</u> $u(\underline{r}_0) = \text{const}$ in $\bar{B}(\underline{r}_0, \delta) \subset \Omega$

Suppose not. Then there is $u(\underline{r}_1) < M$, and since $u \in C^0(\bar{\Omega})$, $u(\underline{r}) < M$ for all r in a $B(\underline{r}_1, \delta_1) \subseteq \bar{B}(\underline{r}_0, \delta)$.

Let $S_2 = \partial B_2(\underline{r}_0, \delta_2)$
where $\delta_2 < \delta$



Then by MVP

$$4\pi\delta_2^2 M = \int_{S_2(\underline{r}_0,\delta_2)} u(\underline{r})\,d\sigma$$

$$= \int_{S_2\cap B_1} u(\underline{r})\,d\sigma + \int_{S_2\cap B_1^c} u(\underline{r})\,d\sigma$$

$$\overset{\text{strict}}{<} \int_{S_2\cap B_1} M\,d\sigma + \int_{S_2\cap B_1^c} M\,d\sigma$$

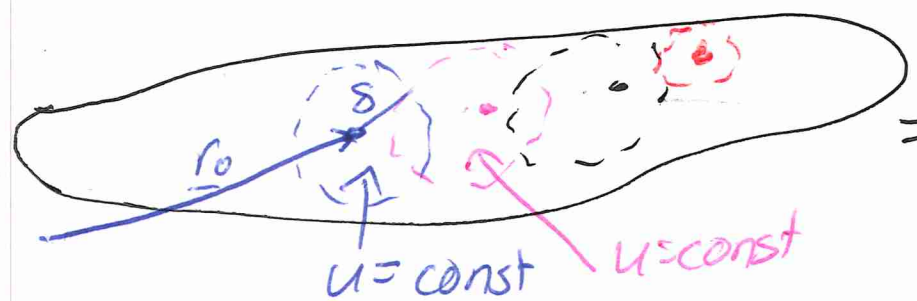
$$= \int_{S_2} M\,d\sigma$$

$$= M\,4\pi\delta_2^2$$

$$4\pi\delta_2^2 M < 4\pi\delta_2^2 M \quad \longrightarrow\longleftarrow$$

$\Rightarrow u(\underline{r}_0) = \text{const}$ in $\bar{B}(\underline{r}_0,\delta)\subset\Omega$

repeat argument



$\Rightarrow u = \text{const}$ in Ω.

$= M$. $\square$

8/8